

Graph Mining Course Project Progress Report

Submission Date: —

Course: Graph Mining [4041]

Instructor: Dr. Zeinab Maleki

Project Title: Graph-Based Sequential App Recommendation (Random Walk, DGNSR, LightGCN)

Student Information

Student Names:

Seyed Ali Elahi

Soroosh Satary

Seyed Mahdi Seyedghaleh

Executive Summary

This project studies graph-based recommendation of mobile applications using three complementary methods: a hybrid Random Walk with Restart (RWR), a Dynamic Graph Neural Network for Sequential Recommendation (DGNSR), and a collaborative-filtering baseline based on LightGCN. The Random Walk model integrates installation-based similarity and co-installation behavior and achieves a top-10 hit rate of 10–12%. DGNSR further improves accuracy to approximately 13–15% in one-hop experiments. Finally, LightGCN was trained on a 50k-user subset of the data, achieving a Hit@10 of about 7–8%, serving as an efficient collaborative baseline.

Progress on Objectives

Objective 1: Graph Construction and Data Modeling

We constructed multiple graph representations, including an app–app similarity graph weighted by installs, a symmetric co-installation graph from user histories, and a user–app bipartite graph for LightGCN.

Objective 2: Random Walk with Restart (RWR)

The hybrid RWR algorithm launches multiple walks from recently installed apps, with restart probability and weighted transitions prioritizing co-install counts and falling back to install similarity.

Objective 3: DGNSR Dynamic Graph Modeling

DGNSR constructs dynamic local subgraphs by alternating between recent items and related users, enabling multi-hop expansion while remaining computationally feasible.

Objective 4: LightGCN Collaborative Filtering

LightGCN learns user and item embeddings over the bipartite graph using normalized adjacency propagation without nonlinearities, trained with BPR loss and negative sampling.

Preliminary Results

Model	Top-10 Hit Rate	Evaluation Notes
Random Walk (RWR)	10–12%	Hybrid co-install + install similarity; ~45k items
DGNSR (1-hop)	13–15%	One-hop expansion; tested on 100k users
LightGCN	7–8%	Trained on 50k-user subset; bipartite user–app graph

Challenges Encountered and Resolutions

Scalability challenges were addressed through distributed preprocessing, neighborhood pruning, and CSR-style indexing. Evaluation consistency across methods is being ensured via a unified Hit@K protocol.

References

1. Huang, Y., Li, T., Wu, Y., Wan, L., Wu, H., Wang, X., & Ning, Z. (2025). An Improved Random Walk Restart Algorithm for Multisimilarity Enhanced Academic Recommendation Systems. *IEEE Transactions on Computational Social Systems*, 12(5), 3103–3115.

2. Li, R., Heitz, L., Inel, O., & Bernstein, A. (2025). D-RDW: Diversity-Driven Random Walks for News Recommender Systems. *Information Processing & Management*, 62(5).

3. TaDGSR: A Time-adaptive Prediction Dynamic Graph Neural Network for Sequential Recommendation. ScienceDirect.